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PI Name:	Hofmann, Douglas Ph.D.		
Project Title:	Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices		
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PI Email:	Douglas.C.Hofmann@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	707-318-5372
Organization Name:	NASA Jet Propulsion Laboratory		
PI Address 1:	4800 Oak Grove Drive		
PI Address 2:	Engineering & Science Directorate, MS 303-400		
PI Web Page:			
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Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
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COI Name (Institution):	Demetriou, Marios (California Institute of Technology) Johnson, William (California Institute of Technology) Rhim, Won-Kyu (California Institute of Technology) Shapiro, Andrew (Jet Propulsion Laboratory)		
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Task Description:	The future materials for NASA spaceflight applications are likely to be high-performance composites. Metallic glass matrix composites, which are metallic glasses reinforced with crystalline dendritic inclusions, were found to impart benchmark properties (see for example, D.C. Hofmann et al. Nature, 451:1085, 2008). The mechanical performance of these composites is understood to be highly dependent on the morphological-feature length scales of the dendritic microstructure. Specifically, it has been demonstrated that the length scales associated with the size and distribution of dendrites has a profound effect on mechanical properties such as ductility, fracture toughness, and fatigue endurance. Understanding and controlling the development of the dendrite microstructure will enable optimization of the mechanical properties and promote a truly unique mechanical performance of these composites. The investigators propose to study the nucleation, growth, directionality and cooling rate dependence of crystalline dendrites evolving in a glass-forming eutectic matrix. The ground-based research effort will be carried out utilizing laboratories at the Jet Propulsion Laboratory and the California Institute of Technology, including the High Vacuum ElectroStatic Levitator platform (HVESL) at Caltech. The investigators propose a four-component ground-based project consisting of (1) directional solidification experiments utilizing Moving Mushy Zone (MMZ) and Bridgman solidification, (2) morphological characterization of dendrite growth and orientation during "mushy" (or "semi-solid") processing, (3) mechanical testing and development of structure-property relations, and (4) dendrite formation and growth during containerless processing utilizing the HVESL. After completion of a ground-based research program, the investigators will extend the experimentation to flight-based microgravity research aboard the ISS, the aim of which is to study gravity-driven convection arising from density gradients in the microstructure. The flight experiment objectives are to (1) use the Low Gradient Furnace (LGF) to complete Bridgman crystal growth on composites in an attempt to grow dendrites along the translational axis, (2) use the Solidification and Quenching Furnace to coarsen and homogenize dendrite microstructures using mushy-zone processing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of innovative new materials is a fundamental part of future terrestrial and space technologies. These materials have a tremendous benefit towards improving life on Earth, from improving the accessibility and cost of solar power, to making cars safer and stronger, to reducing the mass of aircraft, for example. Innovative technologies typically require new materials tailored for those applications. This drives materials development. Metal matrix composites (MMCs) are a relatively new class of metal alloys that exhibit unique properties which make them desirable for many new technologies. MMCs are typically two-phase alloys consisting of a continuous metal matrix incorporated with a second phase (which can be a metal dendrite, a fiber, a particle, or some other inclusion). They differ from traditional monolithic crystalline metal alloys because the mechanical and physical properties of single-phase alloys are limited by composition and dislocation-based plasticity. By creating a composite phase, MMCs have the ability to combine the beneficial properties of both phases into a single alloy, like strength and toughness, for example. A novel type of MMC has been recently developed that exhibits benchmark combinations of strength and toughness. These alloys were developed by creating a MMC with high-strength metallic glasses (also called amorphous metals) as the matrix phase and a soft body centered cubic (bcc) dendrite as the crystalline phase. These metallic glass matrix composites (MGMCs), which are the focus of the current research effort, have been shown to have the strength of carbides, with the toughness and fatigue limit of stainless steel and the density of titanium. These alloys have the potential for significant impact in high-performance structural applications in areas ranging from defense, to aerospace, to commercial sporting goods. The widespread use of MMC (and MGMCs in particular) requires in-depth research into the processing and fabrication of these multi-phase alloys. For example, semi-solid (or mushy) processing is often required to coarsen the second phases in MMCs and MGMCs to improve the mechanical properties. These additional processing requirements make the fabrication of hardware from MMCs unique compared with traditional metal parts (which are typically machined into a final shape). The NASA-funded technology being developed as part of the current program seeks to understand the development of second phases in MGMCs as a function of processing parameters so that innovative industrial processing techniques can be used to generate hardware. The end result is the potential widespread use of MMC in applications such as biomedical implants or spacecraft shielding.
	In the first year of NASA funding we have made significant progress towards the goals of our program outlined in the proposal. Our progress and accomplishments through the end of the first year (October 2010-September 2011) include: (1) the development and study of a new Fe-Ni-based metal matrix composite, (2) electrostatic levitation experiments on metallic glass matrix composites, (3) fabrication, experimentation and testing of a Bridgman-type directional solidification furnace, (4) experimentation and data on moving melt zone processing of metallic glass matrix composites (5) collaborations with other researchers. (1) Development and study of a new Fe-Ni-based metallic glass matrix composite – In addition to the primary objective of the current NASA program (defined as mushy processing of glass-forming matrix composites) we have also been developing new metallic glass matrix composites based in Fe-Ni. Our recent work has shown that we can make FeNi (Invar) dendrites in an FeNiB eutectic and that when rapidly cooled, we can form a glass matrix. This has been achieved by splat quenching microns-thick foils of the alloy from the liquidus. These alloys have interesting potential as cryogenic composites due to the presence of the Invar phase (no thermal expansion) and they are low-cost, be-free composites, which improves their marketability. In this effort, we have collaborated with researchers at UC San Diego to obtain high-resolution TEM micrographs of the composites and X-Ray diffraction scans. Our work has demonstrated a novel composite, based in Fe, and we have also showed control over its microstructure. Next year’s work will involve the preparation and publication of this data in a scientific journal. (2) Electrostatic Levitation Experiments – Using the JPL/Caltech Electrostatic Levitator (ESL) platform located at Caltech, we have successfully completed a first round of experiments on two metal matrix composites that exhibit different volume fractions of second phase (25% and 66%). The ESL offers precise temperature control in a containerless environment, so we were able to perform mushy (or semi-solid) processing on the composites. We first levitated 3 mm diameter spheres of the composite alloys using electrostatic forces. Then, using tetrahedral laser heating, we were able to heat the levitated ingot to any temperature between room temperature and above the liquidus temperature. For our mushy processing experiments, we heated the ingots up to the semi-solid temperature regime, where the liquid coexists with the dendrites, and then held at that temperature isothermally by balancing the laser

Task Progress:	heating with cooling due to radiation. After a prescribed amount of time, the laser heating was turned off and the alloy quenched rapidly due to black-body radiation, allowing us to study the microstructure obtained. We prepared samples of the 67% dendrite-containing alloy after three processing strategies; (i) heating to the liquidus and then cooling, (ii) heating to the liquidus, cooling to 850 °C, holding for 3 min, then cooling, and (iii) heating to the liquidus, cooling to 850 °C, holding for 15 min, then cooling. SEM analysis shows that the dendrites coarsen with isothermal processing, as expected. We were also able to obtain high-temperature data on specific volume, emissivity, cooling curves, recoalescence, liquid-to-liquid phase transitions and melting behavior. Next year, we will obtain high-temperature viscosity data and attempt to correlate the mushy processing time and temperature with the size and morphology of the dendrites through SEM for each alloy. (3) Bridgman-type directional solidification – In the laboratory, we fabricated a furnace used to directionally solidify metallic glass matrix composites. This involved building a graphite oven resting directly above a molten metal bath of Sn-Bi, all inside a vacuum chamber under an inert gas environment. We heated the composites up in the oven using quartz tubes and then slowly lowered the melt through a steep gradient into the metal bath. The gradient from the oven to the bath was from 1000 °C to 120 °C over 1 cm of distance. This allowed us to generate a very steep directional solidification gradient. The alloy in the oven was held in the semi-solid region approximately 1 cm away from the molten metal bath, which was held at a temperature below the glass transition temperature of the composite such that quenching into the bath was sufficient to vitrify the matrix of the composite into a glass. The molten metal bath was necessary for use in a vacuum chamber because of its extremely low vapor pressure. The motivation for this type of processing was to solidify and then grow the dendrites along the gradient of cooling, creating an aligned microstructure. Although the setup worked well, the lack of a backpressure caused extensive porosity in the samples, which prevented tension testing. However, SEM analysis was done on the samples and they showed substantial coarsening of the microstructure, but no directionality. Next year, we are attempting to use a dedicated Bridgman furnace at another location to prepare us porosity-free samples for mechanical testing. We have also been experimenting with semi-solid holding of composite samples using an oven. In contrast to the Bridgman directional solidification, where the samples are quenched along a gradient, the oven experiments are designed for extended semi-solid (or mushy) processing at an isothermal temperature before quenching. This allows the microstructure to coarsen, instead of being directionally aligned. Our aim is to replicate the same processing conditions achieved from electrostatic levitation experiments but in samples large enough for tensile testing. This work will also be our motivation for why low-gravity space flights are required; since density differences between the matrix and the dendrite will cause sedimentation. Next year, we will perform sedimentation experiments to demonstrate this effect. (4) Experimentation and data on moving melt zone processing – Using a water-cooled copper boat, we extensively performed moving melt zone (MMZ) processing on two different metallic glass matrix composites, a Ti-based composite with 40% dendrites and a Zr-Ti-based composite with 66% dendrites. MMZ involves forming a localized melt zone using an RF coil and then translating that zone across the samples in an attempt to create directionality. As with the Bridgman-type solidification, MMZ only heats a localized region of the sample, while the region outside the melt-zone is water-cooled through conduction into the copper boat. SEM micrographs from some early tests shown directionality of the dendrites, but not along the axis of the samples, as expected. Eight samples (4 of each composition) were further subjected to a variety of MMZ heating rates and the samples are currently being prepared for tension testing. Next year, we will attempt to correlate the processing profiles from MMZ with microstructure and mechanical properties through tension and compression testing. (5) As part of this funded program, we have also been working with collaborators to obtain useful data on metallic glass matrix composites that will assist in our understanding of processing. We have been working with researchers at the Ohio State University, UC San Diego and the Indian Institute of Science on these efforts. The major objectives of the second year of NASA funding (October 2011-September 2012) are to correlate mushy processing techniques with mechanical testing, evaluate mushy processing using electrostatic levitation, experiment with sedimentation effects caused by differences in density between the dendrites and the matrix, perform mushy processing using long isothermal holds of composites alloys in an oven, publish papers on our findings, and begin selecting potential candidate alloys for flight experiments.
Bibliography Type:	Description: (Last Updated: 01/01/2026)
Abstracts for Journals and Proceedings	Hofmann DC. "Microgravity Research on Bulk Metallic Glasses and Composites." TMS 2012 Annual Meeting, Orlando FL, March 2012. The Metal Minerals and Materials Society TMS 2012 Annual Meeting, Orlando FL, March 2012. In press, as of August 2011. , Aug-2011